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MICRO-EXPANDED POROUS TRANSPORT LAYERS FOR USE IN ANODE PACK ASSEMBLY AND PROTON EXCHANGE MEMBRANE (PEM) ELECTROLYZERS

Abstract

The present disclosure is directed to an anode porous transport layer (PTL) (**10**), an anode transport layer assembly (**14**), an anode pack assembly (**16**), and a proton exchange membrane electrolyzer device (**18**) comprising two or more micro-expanded metal mesh layers (**12**), wherein each micro-expanded metal mesh layer (**12**) has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation-In-Part of U.S. application Ser. No. 18/776,158 filed Jul. 17, 2024, which claims the benefit of U.S. Provisional Application No. 63/514,262, filed Jul. 18, 2023, the entire contents of which are incorporated herein by reference. [0002] This application also claim the benefit of U.S. Provisional Application No. 63/651,720, filed May 24, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

[0003] The present disclosure generally relates to proton exchange membrane (PEM) electrolyzer devices, and to an anode pack assembly comprising a porous transport layer (PTL) formed from two or more layers of a micro-expanded metal mesh.

[0004] A PEM electrolyzer is composed of a stack, or stacks, of cells each composed of, for example, an anode bipolar plate, an anode flow field, an anode PTL, an anode catalyst layer, a central proton exchange membrane, a cathode catalyst layer, a cathode PTL, a cathode flow field, and a cathode bipolar plate. An example PEM electrolyzer cell is illustrated in FIG. 1. The stack(s) of cells may be secured within an electrolyzer frame and contains a distribution manifold for water inputs and gas outputs to and from each cell.

[0005] Operation of hydrogen electrolyzers includes water circulating through the stack and electrical energy is applied on each side of the anode and cathode. The resulting chemical reaction across the proton exchange membrane splits the water molecule and produces hydrogen and oxygen. That is, water is electrochemically split into hydrogen and oxygen at their respective electrodes, hydrogen at the cathode and oxygen at the anode. PEM water electrolysis is accrued by pumping water to the anode where it is split into oxygen ($\text{O}_{2(g)}$), protons (H^{+}), and electrons (e^{-}). The protons travel via the proton exchange membrane to the cathode side. The electrons exit from the anode through the external power circuit providing the driving force (e.g., cell voltage) for the reaction. At the cathode, the protons and the electrons recombine to provide hydrogen. The reactions are as follows:

Anode: $\text{H}_2\text{O} \rightarrow \frac{1}{2}\text{O}_{2(g)} + 2\text{H}^{+} + 2e^{-}$

Cathode: $2\text{H}^{+} + 2e^{-} \rightarrow \text{H}_2$

Overall: $\text{H}_2\text{O} \rightarrow \text{H}_2 + \frac{1}{2}\text{O}_{2(g)}$

[0006] Catalyst materials are typically applied to the proton exchange membrane. Performance of the electrolyzer is, for example, dependent on the catalyst materials and density, composition and thickness of the proton exchange membrane, electrical contact and conductivity of the various layers, and fluid transport properties within the flow field and PTL.

[0007] The anode PTL of the electrolyzer is typically made of sintered powder, sintered fiber, or metal mesh titanium. The PTL can be a composite of several layers of combinations of powder, fiber, and/or metal mesh with multiple pore sizes.

SUMMARY OF THE DISCLOSURE

[0008] The present disclosure provides an anode PTL of a PEM electrolyzer, comprising: two or

more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.

[0009] The present disclosure further provides an anode transport layer assembly for use in a PEM electrolyzer device comprising: a multilayer anode flow field and an anode PTL comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.

[0010] The present disclosure further provides an anode pack assembly for use in a PEM electrolyzer device comprising: a bipolar plate; an anode flow field; and an anode PTL comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.

[0011] The present disclosure is also directed to a PEM electrolyzer device comprising: a bipolar plate; an anode flow field; an anode PTL comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%; a proton exchange membrane; a cathode electrode; and a cathode PTL.

[0012] The present disclosure further provides a PEM electrolyzer device comprising: a bipolar plate; an anode transport layer assembly comprising a multilayer anode flow field and an anode PTL comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%; a proton exchange membrane; a cathode electrode; and a cathode PTL.

[0013] The present disclosure further provides a PEM electrolyzer device comprising: an anode pack assembly comprising a bipolar plate, an anode flow field, and an anode PTL comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%; a proton exchange membrane; a cathode electrode; and a cathode PTL.

[0014] While exemplary embodiments of the invention have been described as having the features recited, it is understood that various combinations of such features are also encompassed by particular embodiments of the invention and that the scope of the invention is limited by the claims and not the description.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0015] While the specification concludes with claims particularly pointing out and distinctly claiming particular embodiments of the instant invention, various embodiments of the invention can be more readily understood and appreciated from the following descriptions of various embodiments of the invention when read in conjunction with the accompanying drawings.

[0016] FIG. 1 illustrates an exemplary cell construction in a PEM electrolyzer.

[0017] FIG. 2 illustrates an exemplary process wherein micro-expanded metal mesh layers are generated.

[0018] FIG. 3 illustrates an exemplary embodiment of a multilayer micro-expanded metal mesh anode PTL.

[0019] FIG. 4 illustrates an exemplary anode PTL of a PEM electrolyzer including three micro-expanded metal mesh layers;

[0020] FIG. 5 illustrates an exemplary anode transport layer assembly for use in a PEM

electrolyzer device.

[0021] FIG. 6 illustrates an exemplary anode pack assembly for use in a PEM electrolyzer device.

[0022] FIG. 7 illustrates an exemplary PEM electrolyzer device including an anode pack assembly comprising a bipolar plate, a multi-layer anode flow field, and a multilayer micro-expanded metal mesh anode PTL

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0023] Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the device and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present disclosure. Further, in the present disclosure, like-numbered components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-numbered component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape. Further, to the extent that directional terms like top, bottom, up, or down are used, they are not intended to limit the systems, devices, and methods disclosed herein. A person skilled in the art, will recognize that these terms are merely relative to the system and device being discussed and are not universal.

[0024] Unless otherwise specified, when referring to a numerical value, the term “about” is intended to be construed as including a range of values within $\pm 10\%$ of the value being referred to.

[0025] The present disclosure provides for an anode porous transport layer (PTL) **10** (FIGS. 3 and 4) having two or more micro-expanded metal mesh layers **12A**, **12B**, **12C**, . . . **12n**, etc., wherein each micro-expanded metal mesh layer **12A**, . . . **12n**, etc., has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%, an anode porous transport assembly **14** (FIG. 5) including a PTL **10**, an anode pack assembly **16** (FIG. 6) including a PTL **10**, and PEM electrolyzer devices **18** (FIG. 7) comprising a PTL **10**.

[0026] In some embodiments, the PTL **10** comprises two or more micro-expanded metal mesh layers **12A**, **12B**, **12C**, . . . **12n**, etc. to allow for controlling, e.g., thickness, porosity, tortuosity, pore connectivity, and surface roughness while resulting in a near zero distribution in pore sizes. Pore size distribution, or the range of pore sizes existing in a material, is commonly used to characterize random porous materials such as sintered powder and sintered fiber materials, wherein a random porous material will typically exhibit a broad, non-zero distribution. However, the methods described herein for fabricating a micro-expanded metal mesh create uniform pores. Therefore, the micro-expanded metal mesh of the present disclosure, in contrast to random porous materials, has a sharp, near-zero pore size distribution.

Anode Porous Transport Layer (PTL)

[0027] The present disclosure provides, an anode PTL **10** of a PEM electrolyzer **18** comprising: two or more micro-expanded metal mesh layers **12A**, **12B**, . . . **12n**, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.

[0028] As used herein, “micro-expanded” refers to an expanded metal by a process allowing for openings, pores, or slits to be formed into a metal sheet in a predictable and repeatable manner. In

some embodiments, as shown in FIG. 2, a micro-expanded metal mesh layer is formed by, e.g., a slitting blade resulting in a metal that is expanded on a microscopic level creating pores ranging in size from about 3 μm to about 60 μm . In some embodiments, the micro-expanded metal mesh is subsequently flattened through one or more rollers such that the final mesh attains flat surfaces with a predetermined thickness. As used herein, the term “predetermined thickness” refers to a thickness of each micro-expanded metal mesh layer with a defined thickness; that thickness can be adjusted or modified based on, e.g., parameters of the PTL.

[0029] In some embodiments, the purpose of the PTL **10**, sitting between the catalyst layer on the membrane and the flow field in the anode of the cell, is to provide electrical contact to the catalyst layer, remove heat, and/or permit mass transport. The PTL comprises a three-dimensional network of pores within an electronically conductive matrix that provides minimal resistance to fluid transport between the flow field and the catalyst layer and provides intimate electrical contact.

[0030] In some embodiments, the anode PTL **10** comprises two or more micro-expanded metal mesh layers **12**. In some embodiments, the anode PTL **10** comprises two micro-expanded metal mesh layers **12**. In some embodiments, the anode PTL **10** comprises three micro-expanded metal mesh layers **12**. In some embodiments, the anode PTL **10** comprises four micro-expanded metal mesh layers **12**. In some embodiments, the anode PTL **10** comprises five micro-expanded metal mesh layers **12**.

[0031] In some embodiments, the two or more micro-expanded metal mesh layers **12** are oriented in the same direction or oriented in different directions. That is, when the micro-expanded metal mesh layers include pores of an asymmetric shape, having a length along a major axis longer than a length along a minor axis, the two or more micro-expanded metal mesh layers are oriented such that the major axes are oriented parallel to adjacent layers or such that the major axes are oriented 90° to adjacent layers.

[0032] In some embodiments, the metal of each micro-expanded metal mesh layer **12** is titanium, nickel, platinum, ruthenium, iridium, cobalt, zinc, gold, silver, chromium, niobium, zirconium, vanadium, iron, steel, copper, manganese, tungsten, molybdenum, or any alloy thereof. In some embodiments, the metal of each micro-expanded metal mesh layer is the same metal or are different metals of the respective two or more layers. In some embodiments, the metal of the two or more micro-expanded metal mesh layers are titanium.

[0033] In some embodiments, the anode PTL **10** comprises two or more micro-expanded metal mesh layers **12**, wherein the pore size of each micro-expanded metal mesh layer is within the range from about 3 μm to about 100 μm . In some embodiments, the pore size ranges from about 3 μm to about 60 μm . In some embodiments, the pore size ranges from about 10 μm to about 45 μm . In some embodiments, each of the two or more micro-expanded metal mesh layers **12** have different pore sizes. In some embodiments, at least one of the micro-expanded metal mesh layers **12** has a pore size within the range of about 10 μm to about 15 μm . In some embodiments, the PTL comprises three metal mesh layers wherein the pore size of one layer is about 10 μm , one layer is about 20 μm , and one layer is about 40 μm . In some embodiments, two or more micro-expanded metal mesh layers have the same pore size.

[0034] As used herein, the term “pore size” refers to the pore diameter of a circular pore with equivalent open area to the pores of the micro-expanded mesh, which can have a variety of geometric shapes. The “pore size” is measured on the side of the mesh with smaller openings for meshes in which the shape of the slit through the mesh is asymmetric.

[0035] In some embodiments, each micro-expanded metal mesh layer **12** has an open area or porosity ranging from about 10% to about 60%. In some embodiments, the open area or porosity ranges from about 25% to about 55%. In some embodiments, the open area or porosity is about 33%. In some embodiments, the open area or porosity is about 50%. As used herein, the term “open area” or “porosity” means a measure of the void spaces in the micro-expanded mesh layer and is expressed as a fraction of the projected area of voids over the total projected area of the mesh or as

an equivalent percentage such as from 10% to 60%. In some embodiments, each micro-expanded metal mesh layer **12** has an open area ranging from about 20% to about 60%. In some embodiments, each micro-expanded metal mesh layer **12** of the anode porous transport layer **10** has an open area with the same percentage or a different percentage of the respective other layers. In some embodiments, the “open area” or “porosity” is determined by a camera system such as an infrared (IR) thermal camera, a high-speed vision camera, or two or more cameras.

[0036] In some embodiments, each micro-expanded metal mesh layer **12** has a thickness ranging from about 5 μm to about 100 μm . In some embodiments, the thickness ranges from about 10 μm to about 60 μm . In some embodiments, each of the two or more micro-expanded metal mesh layers **12** has the same thickness or a different thickness.

[0037] In some embodiments, each micro-expanded metal mesh layer **12** has a through-plane tortuosity ranging from about 1 to about 3. As used herein, through-plane tortuosity refers to the ratio of the effective path length of a particle across the plane of the PTL **10** to the thickness of the PTL. In some embodiments, tortuosity is estimated with the arc-chord method using a 3-D model of the PTL. In some embodiments, the tortuosity ranges from 1 to about 2.3.

[0038] In some embodiments, the face of the PTL **10** in contact with the proton exchange membrane has a surface roughness ranging from about 25 nm to about 10 μm . As used herein, “surface roughness” refers to the arithmetic average of profile height deviations from the mean line, known generally as Ra. In some embodiments, surface roughness is measured by profilometry. In some embodiments, the surface roughness ranges from about 25 nm to about 1 μm . In some embodiments, the surface roughness ranges from about 25 nm to about 500 nm. In some embodiments, the surface roughness ranges from about 25 nm to about 100 nm.

Anode Transport Layer Assembly

[0039] Referring to FIG. 5, the present disclosure further provides an anode transport layer assembly **14** for use in a PEM electrolyzer device **18** comprising: a multilayer anode flow field **20**; and an anode PTL **10** comprising two or more micro-expanded metal mesh layers **12**, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.

[0040] As used herein, a flow field **20**, whether anode or cathode, is generally defined as a porous, electronically conductive component intended to transport liquid and gas within the plane of the flow field into or out of the cell while also providing electronic transport across the layer from the PTL **10** to a bipolar plate. As used herein, a multilayer flow field **20** is generally defined as a plurality of porous metal layers that are stacked in such a way to form a 3-dimensional network of pores permitting fluid flow within the plane of the flow field. For example, the multilayer anode flow field **20** may comprise a plurality of metal layers of larger porosity with a pore range of about 0.3 mm to about 6 mm. In some embodiments, some or all of the flow field layers may comprise expanded metal layers.

[0041] In some embodiments, the flow field may comprise a single porous layer.

Anode Pack Assembly

[0042] Referring now to FIG. 6, the present disclosure further provides an anode pack assembly **16** for use in a PEM electrolyzer device **18** comprising: a bipolar plate **22**; an anode flow field **20**; and an anode PTL **10** comprising two or more micro-expanded metal mesh layers **12**, wherein each micro-expanded metal mesh layer **12** has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.

[0043] For example, according to FIG. 6, the anode bipolar plate **22** is conventional in the art, such as a solid, flat plate of titanium extending beyond the area of the flow field **20** and anode PTL **10** with openings in these external edges connecting the cell to external bulk fluid flow into and out of the cell.

[0044] In some embodiments, the anode flow field **20** is a multilayer flow field comprised of a plurality of porous metal layers that are stacked in such a way to form a 3-dimensional network of

pores permitting fluid flow within the plane of the flow field. For example, the multilayer anode flow field **20** may comprise a plurality of metal layers of larger porosity with a pore range of about 0.3 mm to about 6 mm. In some embodiments, some or all of the flow field layers may comprise expanded metal mesh layers with a pore range of about 0.3 mm to about 6 mm. In some embodiments, the multilayer flow field may comprise expanded metal layers. The multilayer PTL **10** may comprise a plurality of layers of micro-expanded mesh titanium foil. The individual layers may have varying pore sizes as will be described hereinafter. In some embodiments, pore sizes may range from about 3 μm to about 60 μm and the expanded metal foil layers may have a mesh open area of about 10% to about 60%. In some embodiments there may be three micro-expanded metal layers of decreasing pore size traversing from the flow field side to the catalyst side.

[0045] In some embodiments, the flow field may comprise a single porous layer.

[0046] As used herein, a bipolar plate **22** is generally defined as an electronically conducting plate that does not permit liquid or gas transport through the plate within the electrochemically active area of the cell. Functionally, the bipolar plate separates the reactants and products of adjacent electrochemical cells while electronically connecting the cells in series to form the stack. A person skilled in the art will recognize that the design of the bipolar plate can vary widely. A typical bipolar plate will be composed of either metal or graphite and may have holes on the sides of the plates used to create channels, termed an internal manifold, through the stack for the distribution of water into the cells and water and gas products out of the cells. A bipolar plate **22** may or may not feature channels patterned onto the surfaces of the plate used to direct the flow of water and gases across the cell, and may or may not have channels interior to the plate used for coolant flow.

[0047] A person skilled in the art will recognize that the use of the term “bipolar” plate is conventional and can be extended to apply to the first and final such plates in the stack or both plates of a stack of one cell. Such plates are properly termed “monopolar” plates because they do not connect opposite electrodes of two adjacent cells but are functionally identical to bipolar plates on the side in which they are in contact with a cell.

[0048] In some embodiments, the anode flow field **20** and the bipolar plate **22** are combined into a single component in which the anode flow field is comprised of channels built into the surface of the bipolar plate.

Proton Exchange Membrane (PEM) Electrolyzer Device

[0049] The present disclosure is also directed to a PEM electrolyzer device **18** comprising: a bipolar plate **22**; an anode flow field **20**; an anode PTL **10** comprising two or more micro-expanded metal mesh layers **12**, wherein each micro-expanded metal mesh layer **12** has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%; an anode catalyst layer **24**; a proton exchange membrane **26**; a cathode catalyst layer **28**; a cathode PTL **30**; a cathode flow field **32**; and another bipolar plate **34**. The anode/cathode stack assembly **36** is captured within a frame structure **38** including suitable water inputs to feed the device and gas outputs to capture the output gases.

[0050] The present disclosure further provides a PEM electrolyzer device **18** comprising: a bipolar plate **22**; an anode transport layer assembly **14** as defined above; an anode catalyst layer **24**; a proton exchange membrane **26**; a cathode catalyst layer **28**; a cathode PTL **30**; a cathode flow field **32**; and another bipolar plate **34**.

[0051] The present disclosure further provides a PEM electrolyzer device **18** comprising: an anode pack assembly **16** as defined above; an anode catalyst layer **24**; a proton exchange membrane **26**; a cathode catalyst layer **28**; a cathode PTL **30**; a cathode flow field **32**; and another bipolar plate **34**.

[0052] As used herein, “catalyst layer,” in the context of the anode side of the electrolyzer, will be understood by one of ordinary skill in the art to refer to a layer composed of a material that catalyzes the oxidation of water to oxygen with concomitant release of protons and electrons. Likewise, “catalyst layer,” in the context of the cathode side of the electrolyzer, will be understood by one of ordinary skill in the art to refer to a layer composed of a material that catalyzes the

reduction of protons to hydrogen with the use of electrons. As non-limiting examples, the anode catalyst layer **24** may be composed of iridium oxide while the cathode catalyst layer **28** may be composed of platinum. The catalyst layers are typically adsorbed onto either the proton exchange membrane **26** or the PTL **10/30**. The catalyst layers **24, 28** may also include non-catalytic materials that enhance proton or electron conductivity within the layers.

[0053] As used herein, “proton exchange membrane” will be understood by one of ordinary skill in the art to refer to a layer **26** having the function of conducting protons through the membrane while preventing both bulk fluid flow through the membrane and electron flow across the membrane. As a non-limiting example, the proton exchange membrane **26** may be a 50-200 μm thick layer of perfluorosulfonic acid polymer.

[0054] As used herein, the term “cathode PTL” is generally defined as the component on the cathode side of the cell that has the same functions as the anode PTL. The cathode PTL **30** is typically composed of carbon-based materials such as carbon cloth, carbon felt, or carbon paper.

[0055] As used herein, the term “cathode flow field” is generally defined as the component on the cathode side of the cell that has the same functions as the anode flow field. The cathode flow field **32** is typically composed of carbon-based materials such as carbon cloth, carbon felt, or carbon paper or is built into the bipolar plate as channels.

Solid-State Bonding

[0056] In some embodiments, a drawback of expanded metal flow fields in PEM electrolyzer systems can be the high contact resistance that develops between adjacent expanded mesh layers during operation. For example, in titanium-based expanded mesh, this high contact resistance is attributed to the growth of a resistive layer of TiO_2 . Methods to prevent this increase in interfacial contact resistance are limited and current practice is to coat each individual mesh layer with an expensive corrosion resistant coating. Other options to prevent the increase in contact resistance include, for example, solid-state bonding. Herein, solid-state bonding will refer to the process of bonding metals in intimate contact using high pressure and temperature. With solid-state bonding, adjacent mesh layers of the two or more micro-expanded metal mesh layers **12** are metallurgically bonded to each other. Solid-state bonding may eliminate the need for corrosion resistant coatings of individual mesh layers **12**.

[0057] In some embodiments, solid-state bonding of titanium, e.g., the two or more micro-expanded metal mesh layers **12**, is carried out at high pressure (such as from about 0.1 MPa to about 10 MPa), high temperature (such as from about 700° C. to about 1100° C.), and within high vacuum (such as from about 10^{sup.}-6 torr to about 10^{sup.}-3 torr) or under high purity inert atmosphere (such as 99.999% argon). Given these parameters, specialized vacuum furnaces and fixturing capable of applying the requisite pressures in-situ may be utilized to achieve the high pressure, high temperature, and high vacuum.

[0058] Further, under these conditions, atoms inter-diffuse across the mating surfaces of the work pieces to form a monolithic joint, eliminating the surface interface. The partial or whole elimination of surface interfaces limits the growth of TiO_2 , thereby reducing the interfacial contact resistance. Solid-state bonding can therefore bond the individual mesh layers of the flow field together.

[0059] In some embodiments, solid-state bonding can bond the outermost layers of the flow field to the PTL **10** and to the bipolar plate **22**. In some embodiments, for a micro-expanded PTL **10**, solid-state bonding can also bond the multiple layers **12** of micro-expanded mesh to form the PTL **10**. By doing so, in some embodiments, bonding occurs in all the layers of the PTL **10**, the flow field **20**, and the bipolar plate **22** to generate an integrated anode pack assembly **16** in one step. Similarly, in some embodiments, solid-state bonding can bond the layers **12** of the PTL **10** and the flow field **20** to generate an integrated anode transport layer assembly **14**.

[0060] In some embodiments, the implementation of solid-state bonding in fabrication of the anode pack **16** can improve the interfaces between the PTL **10**, flow field **20**, and bipolar plate **22** by the

creation of metallurgical bonds between adjoining layers. With the improved bonding, the anode pack **16** has improved durability without the need for precious metal corrosion inhibitors within the interior of the anode pack **16** (i.e., without coating each individual later).

[0061] While there is shown and described herein certain specific structure embodying the present disclosure, it will be manifest to those skilled in the art that various modifications and rearrangements of the parts may be made without departing from the spirit and scope of the underlying inventive concept and that the same is not limited to the particular forms herein shown and described.

[0062] Without limitation, some embodiments of the present disclosure include the following:

Exemplary Anode Porous Transport Layers

[0063] Embodiment 1A: An anode porous transport layer, comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%. [0064]

Embodiment 1B: The porous transport layer according to Embodiment 1A, wherein the micro-expanded metal mesh layers comprise three to five layers. [0065]

Embodiment 1C: The porous transport layer according to any of Embodiments 1A-1B, wherein each micro-expanded metal mesh layer is titanium, nickel, platinum, ruthenium, iridium, cobalt, zinc, gold, silver, chromium, niobium, zirconium, vanadium, iron, steel, copper, manganese, tungsten, molybdenum, an alloy thereof, or any combination thereof. [0066]

Embodiment 1D: The porous transport layer according to any of Embodiments 1A-1C, wherein each of the two or more micro-expanded metal mesh layers has the same pore size. [0067]

Embodiment 1E: The porous transport layer according to any of Embodiments 1A-1D, wherein each of the two or more micro-expanded metal mesh layers have a different pore size. [0068]

Embodiment 1F: The porous transport layer according to any of Embodiments 1A-1E, wherein the micro-expanded metal mesh layers comprise three layers and each of the three micro-expanded metal mesh layers has a different pore size ranging from about 3 μm to about 60 μm . [0069]

Embodiment 1G: The porous transport layer according to any of Embodiments 1A-1F, wherein the first micro-expanded metal mesh layer has a 12.5 μm thickness with 10 μm pores, the second micro-expanded metal mesh layer has a 25 μm thickness with 20 μm pores, and the third micro-expanded metal mesh layer has a 60 μm thickness with 45 μm pores. [0070]

Embodiment 1H: The porous transport layer according to any of Embodiments 1A-1G, wherein the two or more micro-expanded metal mesh layers have an open area greater than or equal to 20% to about 60%. [0071]

Embodiment 1I: The porous transport layer according to any of Embodiments 1A-1H, wherein the open area of the two or more micro-expanded metal mesh layers have range from about 25% to about 55%. [0072]

Embodiment 1J: The porous transport layer according to any of Embodiments 1A-1I, wherein the two or more micro-expanded metal mesh layers have a pore size ranging from about 10 μm to about 15 μm . [0073]

Embodiment 1K: The porous transport layer according to any of Embodiments 1A-1J, wherein each micro-expanded metal mesh layer has a thickness ranging from about 5 μm to about 100 μm . [0074]

Embodiment 1L: The porous transport layer according to any of Embodiments 1A-1K, wherein each of the two or more micro-expanded metal mesh layers has the same thickness or different thickness. [0075]

Embodiment 1M: The porous transport layer according to any of Embodiments 1A-1L, wherein each of the two or more micro-expanded metal mesh layers has a thickness ranging from about 10 μm to about 60 μm . [0076]

Embodiment 1N: The porous transport layer according to any of Embodiments 1A-1M, wherein each of the two or more micro-expanded metal mesh layers has a tortuosity ranging from 1 to about 2.3. [0077]

Embodiment 1O: The porous transport layer according to any of Embodiments 1A-1N, wherein the surface roughness of the surface of the micro-expanded mesh in contact with the proton exchange membrane is within the range of 25 to 100 nm. [0078]

Embodiment 1P: The porous transport layer according to any of Embodiments 1A-1O, wherein adjacent micro-expanded mesh layers are oriented 90° to one another.

Exemplary Anode Transport Layer Assemblies

[0079] Embodiment 2A: An anode transport layer assembly for use in a PEM electrolyzer device comprising: a multilayer anode flow field; and an anode porous transport layer comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 30 μm and an open area ranging from about 10% to about 60%. [0080] Embodiment 2B: The anode transport layer assembly according to Embodiment 2A, wherein the micro-expanded metal mesh layers comprise three to five layers. [0081] Embodiment 2C: The anode transport layer assembly according to any of Embodiments 2A-2B, wherein each micro-expanded metal mesh layer is titanium, nickel, platinum, ruthenium, iridium, cobalt, zinc, gold, silver, chromium, niobium, zirconium, vanadium, iron, steel, copper, manganese, tungsten, molybdenum, an alloy thereof, or any combination thereof. [0082] Embodiment 2D: The anode transport layer assembly according to any of Embodiments 2A-2C, wherein each of the two or more micro-expanded metal mesh layers has the same pore size. [0083] Embodiment 2E: The anode transport layer assembly according to any of Embodiments 2A-2D, wherein each of the two or more micro-expanded metal mesh layers have a different pore size. [0084] Embodiment 2F: The anode transport layer assembly according to any of Embodiments 2A-2E, wherein the micro-expanded metal mesh layers comprise three layers, and wherein the first micro-expanded metal mesh layer has a 12.5 μm thickness with 10 μm pores, the second micro-expanded metal mesh layer has a 25 μm thickness with 20 μm pores, and the third micro-expanded metal mesh layer has a 60 μm thickness with 45 μm pores. [0085] Embodiment 2G: The anode transport layer assembly according to any of Embodiments 2A-2F, wherein each of the two or more micro-expanded metal mesh layers has a tortuosity ranging from 1 to about 2.3. [0086] Embodiment 2H: The anode transport layer assembly according to any of Embodiments 2A-2G, wherein the surface roughness of the surface of the micro-expanded mesh in contact with the proton exchange membrane is within the range of 25 to 100 nm. [0087] Embodiment 2I: The anode transport layer assembly according to any of Embodiments 2A-2H, wherein adjacent micro-expanded mesh layers are oriented 90° to one another. [0088] Embodiment 2J: The anode transport layer assembly according to any of Embodiments 2A-2I, wherein the multilayer anode flow field comprises a plurality of mesh metal layers with a pore range of about 0.3 mm to about 6 mm. [0089] Embodiment 2K: The anode transport layer assembly according to any of Embodiments 2A-2J, wherein the plurality of mesh metal layers comprise expanded metal layers.

Exemplary Anode Pack Assemblies

[0090] Embodiment 3A: An anode pack assembly for use in a PEM electrolyzer device comprising: a bipolar plate; an anode flow field; and an anode porous transport layer comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 30 μm and an open area ranging from about 10% to about 60%. [0091] Embodiment 3B: The anode pack assembly according to Embodiment 3A, wherein the micro-expanded metal mesh layers comprise three to five layers. [0092] Embodiment 3C: The anode pack assembly according to any of Embodiments 3A-3B, wherein each micro-expanded metal mesh layer is titanium, nickel, platinum, ruthenium, iridium, cobalt, zinc, gold, silver, chromium, niobium, zirconium, vanadium, iron, steel, copper, manganese, tungsten, molybdenum, an alloy thereof, or any combination thereof. [0093] Embodiment 3D: The anode pack assembly according to any of Embodiments 3A-3C, wherein each of the two or more micro-expanded metal mesh layers has the same pore size. [0094] Embodiment 3E: The anode pack assembly according to any of Embodiments 3A-3D, wherein each of the two or more micro-expanded metal mesh layers have a different pore size. [0095] Embodiment 3F: The anode pack assembly according to any of Embodiments 3A-3E, wherein the micro-expanded metal mesh layers comprise three layers, and wherein the first micro-expanded metal mesh layer has a 12.5 μm thickness with 10 μm pores, the second micro-expanded metal mesh layer has a 25 μm thickness with 20 μm pores, and the third micro-expanded metal mesh layer has a 60 μm thickness with 45 μm pores. [0096] Embodiment 3G: The anode pack assembly according to any of Embodiments 3A-3F, wherein each of the two or

more micro-expanded metal mesh layers has a tortuosity ranging from 1 to about 2.3. [0097] Embodiment 3H: The anode pack assembly according to any of Embodiments 3A-3G, wherein the surface roughness of the surface of the micro-expanded mesh in contact with the proton exchange membrane is within the range of 25 to 100 nm. [0098] Embodiment 3I: The anode pack assembly according to any of Embodiments 3A-3H, wherein adjacent micro-expanded mesh layers are oriented 90° to one another. [0099] Embodiment 3J: The anode pack assembly according to any of Embodiments 3A-3I, wherein the anode flow field comprises a plurality of mesh metal layers with a pore range of about 0.3 mm to about 6 mm. [0100] Embodiment 3K: The anode pack assembly according to any of Embodiments 3A-3J, wherein the plurality of mesh metal layers comprise expanded metal layers. [0101] Embodiment 3L: The anode pack assembly according to any of Embodiments 3A-3K, wherein the bipolar plate comprises an electronically conducting material and wherein the bipolar plate does not permit liquid or gas transport across the plate. [0102] Embodiment 3M: The anode pack assembly according to any of Embodiments 3A-3L, wherein the bipolar plate comprises metal or graphite. [0103] Embodiment 3N: The anode pack assembly according to any of Embodiments 3A-3M, wherein the bipolar plate includes channels patterned onto the plate to direct the flow of water and gases.

Exemplary PEM Electrolyzer Devices

[0104] Embodiment 4A: A PEM electrolyzer device comprising: a bipolar plate; an anode flow field; an anode porous transport layer comprising a porous transport layer comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 30 μm and an open area ranging from about 10% to about 60%; an anode catalyst layer; a proton exchange membrane; a cathode catalyst layer; a cathode porous transport layer; and a cathode flow field. [0105] Embodiment 4B: The PEM electrolyzer device according to Embodiment 4A, wherein the micro-expanded metal mesh layers comprise three to five layers. [0106] Embodiment 4C: The PEM electrolyzer device according to any of Embodiments 4A-4B, wherein each micro-expanded metal mesh layer is titanium, nickel, platinum, ruthenium, iridium, cobalt, zinc, gold, silver, chromium, niobium, zirconium, vanadium, iron, steel, copper, manganese, tungsten, molybdenum, an alloy thereof, or any combination thereof. [0107] Embodiment 4D: The PEM electrolyzer device according to any of Embodiments 4A-4C, wherein each of the two or more micro-expanded metal mesh layers has the same pore size. [0108] Embodiment 4E: The PEM electrolyzer device according to any of Embodiments 4A-4D, wherein each of the two or more micro-expanded metal mesh layers have a different pore size. [0109] Embodiment 4F: The PEM electrolyzer device according to any of Embodiments 4A-4E, wherein the micro-expanded metal mesh layers comprise three layers, and wherein the first micro-expanded metal mesh layer has a 12.5 μm thickness with 10 μm pores, the second micro-expanded metal mesh layer has a 25 μm thickness with 20 μm pores, and the third micro-expanded metal mesh layer has a 60 μm thickness with 45 μm pores. [0110] Embodiment 4G: The PEM electrolyzer device according to any of Embodiments 4A-4F, wherein each of the two or more micro-expanded metal mesh layers has a tortuosity ranging from 1 to about 2.3. [0111] Embodiment 4H: The PEM electrolyzer device according to any of Embodiments 4A-4G, wherein the surface roughness of the surface of the micro-expanded mesh in contact with the proton exchange membrane is within the range of 25 to 100 nm. [0112] Embodiment 4I: The PEM electrolyzer device according to any of Embodiments 4A-4H, wherein adjacent micro-expanded mesh layers are oriented 90° to one another. [0113] Embodiment 4J: The PEM electrolyzer device according to any of Embodiments 4A-4I, wherein the anode flow field comprises a plurality of mesh metal layers with a pore range of about 0.3 mm to about 6 mm. [0114] Embodiment 4K: The PEM electrolyzer device according to any of Embodiments 4A-4J, wherein the plurality of mesh metal layers comprise expanded metal layers. [0115] Embodiment 4L: The PEM electrolyzer device according to any of Embodiments 4A-4K, wherein the bipolar plate comprises an electronically conducting material and wherein the bipolar plate does not permit liquid or gas transport across the plate. [0116] Embodiment 4M: The

PEM electrolyzer device according to any of Embodiments 4A-4L, wherein the bipolar plate comprises metal or graphite. [0117] Embodiment 4N: The PEM electrolyzer device according to any of Embodiments 4A-4M, wherein the bipolar plate includes channels patterned onto the plate to direct the flow of water and gases. [0118] Embodiment 4O: The PEM electrolyzer device according to any of Embodiments 4A-4N, wherein the anode catalyst layer comprises iridium oxide. [0119] Embodiment 4P: The PEM electrolyzer device according to any of Embodiments 4A-4O, wherein the cathode catalyst layer comprises platinum. [0120] Embodiment 4Q: The PEM electrolyzer device according to any of Embodiments 4A-4P, wherein the proton exchange membrane comprises a 50-200 μm thick layer of perfluorosulfonic acid polymer. [0121] Embodiment 4R: The PEM electrolyzer device according to any of Embodiments 4A-4Q, wherein the cathode porous transport layer and the cathode flow field comprise a carbon-based material. [0122] Embodiment 4S: The PEM electrolyzer device according to any of Embodiments 4A-4R, wherein the carbon-based material includes carbon cloth, carbon felt, or carbon paper. [0123] Embodiment 4T: The PEM electrolyzer device according to any of Embodiments 4A-4S, wherein the cathode flow field is built into the bipolar plate as channels.

EXAMPLES

Example 1: Anode Porous Transport Layer (PTL)

[0124] In an exemplary embodiment, an anode porous transport layer of a proton exchange membrane includes three micro-expanded metal mesh layers as shown in FIG. 4. In FIG. 4, a model of a 540 μm ×540 μm section of a PTL comprising three layers of micro-expanded mesh in exploded view and combined. Pore sizes of each layer are 10 μm , 20 μm , and 45 μm such that a fluid path exists through the PTL for each 10 μm pore.

Claims

1. An anode porous transport layer, comprising: two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 60 μm and an open area ranging from about 10% to about 60%.
2. The porous transport layer according to claim 1, wherein the micro-expanded metal mesh layers comprise three to five layers.
3. The porous transport layer according to claim 1, wherein each of the two or more micro-expanded metal mesh layers has the same pore size or different pore size.
4. The porous transport layer according to claim 2, wherein the micro-expanded metal mesh layers comprise three layers and each of the three micro-expanded metal mesh layers has a different pore size ranging from about 3 μm to about 60 μm .
5. The porous transport layer according to claim 4, wherein the first micro-expanded metal mesh layer has a 12.5 μm thickness with 10 μm pores, the second micro-expanded metal mesh layer has a 25 μm thickness with 20 μm pores, and the third micro-expanded metal mesh layer has a 60 μm thickness with 45 μm pores.
6. The porous transport layer according to claim 1, wherein the two or more micro-expanded metal mesh layers have an open area greater than or equal to 20% to about 60%.
7. The porous transport layer according to claim 1, wherein each micro-expanded metal mesh layer has a thickness ranging from about 5 μm to about 100 μm .
8. The porous transport layer according to claim 7, wherein each of the two or more micro-expanded metal mesh layers has the same thickness or different thickness.
9. The porous transport layer according to claim 1, wherein each of the two or more micro-expanded metal mesh layers has a tortuosity ranging from 1 to about 2.3.
10. The porous transport layer according to claim 1, wherein the surface roughness of the surface of the micro-expanded mesh in contact with the proton exchange membrane is within the range of 25 to 100 nm.

- 11.** The porous transport layer according to claim 1, wherein adjacent micro-expanded mesh layers are oriented 90° to one another.
 - 12.** An anode transport layer assembly for use in a PEM electrolyzer device comprising: a multilayer anode flow field; and an anode porous transport layer comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 30 μm and an open area ranging from about 10% to about 60%.
 - 13.** The anode transport layer assembly according to claim 12, wherein each of the two or more micro-expanded metal mesh layers has the same pore size or different pore size.
 - 14.** The anode transport layer assembly according to claim 13, wherein the micro-expanded metal mesh layers comprise three layers, and wherein the first micro-expanded metal mesh layer has a 12.5 μm thickness with 10 μm pores, the second micro-expanded metal mesh layer has a 25 μm thickness with 20 μm pores, and the third micro-expanded metal mesh layer has a 60 μm thickness with 45 μm pores.
 - 15.** The anode transport layer assembly according to claim 12, wherein each of the two or more micro-expanded metal mesh layers has a tortuosity ranging from 1 to about 2.3.
 - 16.** The anode transport layer assembly according to claim 12, wherein the surface roughness of the surface of the micro-expanded mesh in contact with the proton exchange membrane is within the range of 25 to 100 nm.
 - 17.** The anode transport layer assembly according to claim 12, wherein adjacent micro-expanded mesh layers are oriented 90° to one another.
 - 18.** An anode pack assembly for use in a PEM electrolyzer device comprising: a bipolar plate; an anode flow field; and an anode porous transport layer comprising two or more micro-expanded metal mesh layers, wherein each micro-expanded metal mesh layer has a pore size ranging from about 3 μm to about 30 μm and an open area ranging from about 10% to about 60%.
 - 19.** The anode pack assembly according to claim 18, wherein the micro-expanded metal mesh layers comprise three to five layers.
 - 20.** The anode pack assembly according to claim 18, wherein each of the two or more micro-expanded metal mesh layers has the same pore size.
 - 21.** The anode pack assembly according to claim 18, wherein each of the two or more micro-expanded metal mesh layers have a different pore size.
 - 22.** The anode pack assembly according to claim 21, wherein the micro-expanded metal mesh layers comprise three layers, and wherein the first micro-expanded metal mesh layer has a 12.5 μm thickness with 10 μm pores, the second micro-expanded metal mesh layer has a 25 μm thickness with 20 μm pores, and the third micro-expanded metal mesh layer has a 60 μm thickness with 45 μm pores.
 - 23.** The anode pack assembly according to claim 18, wherein each of the two or more micro-expanded metal mesh layers has a tortuosity ranging from 1 to about 2.3.
 - 24.** The anode pack assembly according to claim 18, wherein the surface roughness of the surface of the micro-expanded mesh in contact with the proton exchange membrane is within the range of 25 to 100 nm.
 - 25.** The anode pack assembly according to claim 18, wherein adjacent micro-expanded mesh layers are oriented 90° to one another.
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